



Design and Implementation of Physical Education Teaching Management System Based on Multi-agent Model

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Abstract

The traditional physical education (PE) teaching management system is usually controlled and managed by a single center, which cannot meet the diversified and personalized teaching needs. Therefore, the research of PE teaching management system based on multi-agent mode has become an important direction. The purpose of this paper was to discuss how to improve the effect and quality of PE teaching and enhance students' learning enthusiasm and initiative through the design of multi-agent mode PE teaching management system. The PE teaching management system based on multi-agent mode provides more flexible and personalized teaching management services by utilizing the cooperation and interaction between agents, realizes the information exchange between teachers and students, provides real-time teaching feedback and evaluation, and promotes the sharing and collaboration of teaching resources. Therefore, the operating efficiency of the conventional physical education management system was the highest at 75% and the lowest at 67%, according to the experimental findings of this paper. The multi-agent model-based management system for physical education had a 95 percent maximum operating efficiency and an 88% minimum operational efficiency. The minimum difference between the two was 21%, and the maximum difference was 20%. It can be seen that the operation efficiency of the physical education management system based on the multi-agent model is more advantageous and more stable.

Keywords Physical education · Multi-agent model · Management system · Online teaching · Software

1 Introduction

Traditional systems are often controlled by the center and lack of interaction and communication between teachers and students. However, the multi-agent mode of PE teaching management system emphasizes the cooperation and interaction between agents, and realizes the centralized management of teaching resources in the traditional system of teachers and students, which lacks flexibility and sharing. In the multi-agent mode, distributed teaching management and resource sharing can be realized, and teachers and students can jointly manage and use teaching resources to optimize the teaching process. Mutual communication between students provides better teaching feedback and guidance. The

design and implementation of PE teaching management system based on multi-agent mode aims to provide a more flexible, personalized and cooperative teaching environment, which is helpful to improve the quality and effect of PE teaching and meet the different needs of students. At present, with the comprehensive promotion of the national fitness program, people's concept of sports and training has changed fundamentally [1]. However, based on different age levels, different sports bases and goal pursuits, various groups of people have different needs for sports training content, methods and items, and even diametrically different [2]. Therefore, it is necessary to investigate and analyze different needs one by one, and give them personalized training guidance. The traditional manual management decision-making mode obviously cannot be adapted. It follows the decision-making concept and method based on the subjective experience of managers, and cannot formulate scientific training methods and provide targeted training items for the differences of individual sports ability levels of different training objects. This not only inhibits people's enthusiasm for sports training, but also directly affects the effect of sports training.

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Therefore, it is urgent to reform the decision-making mode of sports training management.

As an important part of the all-round development of “morality, intelligence, physique, beauty, and work”, physical education has begun to be valued by colleges and universities across the country, and some areas have directly incorporated physical education into the subject of student evaluation [3]. The implementation of physical education management plan is often divided into many steps, which are jointly carried out by many departments. To make these steps connect with each other and the departments cooperate with each other, managers need to control the work process. With the establishment and improvement of the school network platform, most students share the benefits and value of network teaching, and the realization of network teaching in physical education has gradually become the common goal of colleges and universities, teachers and students [4]. With the continuous reform of the educational system and the expansion of the school scale, the number of students has increased, and the teaching resources have become more and more limited. Given the large number of students and teachers, the main problem facing schools is how to manage their information resources and make full use of their learning resources. Only by adapting to the development of the school, the teaching management system can be flexible, reliable, and conservative. The application efficiency of Agent technology in intelligent decision management system is extremely prominent. It can interact with users in an appropriate way, make action plans and content independently according to the previous decision content, and assist in intelligent management and decision-making. Therefore, the use of multi-agent mode to carry out data analysis, personalized recommendation and decision management in a collaborative manner has become a new direction in the innovative design of current sports training management system.

Network sports management greatly improves the efficiency and effect of school sports work, which is a huge change to the traditional model and provides a solid foundation for colleges and universities to implement network sports. Online teaching research in other countries is relatively mature, and the development of online teaching in China has begun to take shape [5]. Many schools have established online teaching management systems. However, with the advancement of technology, the drawbacks of traditional software development methods begin to appear. Multi-agent technology originated from the research of distributed artificial intelligence. It has developed rapidly since it appeared in the 1970s. Now it has become a thinking method and tool for analyzing and simulating complex systems. At present, many large-scale software have adopted intelligent agent technology, which shows that intelligence is the new direction of software design. The research content and innovation of this

paper is that the intelligent agent technology is introduced into the traditional learning management system, which has improved the efficiency and performance of the system.

1. It is assumed that the cooperation and interaction between agents can effectively improve the teaching effect and quality.
2. It is assumed that individualized and adaptive teaching management can meet the needs of different students.
3. It is assumed that distributed management and resource sharing can optimize the teaching process.

Limitations of the design of PE teaching management system based on multi-agent model:

1. System complexity: the system design based on the multi-agent model may increase the complexity of the system, including the communication, cooperation and conflict resolution between agents. This can lead to high system implementation and maintenance costs.
2. Technical limitations: the multi-agent model needs to support powerful computing and communication capabilities so that agents can interact with each other in real time and process large amounts of data. In some technical conditions limited environment, there may be some technical challenges.

2 Related Works

The teaching management system is a professional management platform that realizes the daily business management of teachers and the comprehensive performance evaluation of students. It is the focus of school work, and its quality is related to the construction of the whole school. Fan [6] proposed a design of a sports teaching management system based on web technology to address the current issue of incomplete construction of sports teaching management systems in most universities. In the original hardware system structure, an automatic reset circuit was added to detect circuit status and ensure charging stability. The outflow of resource information was avoided to reduce interdependence between modules. Web technology was utilized to improve the functional structure of system management. The fitness function was utilized, and the course management plan was refined to complete the design of the physical education teaching management system. Using Pingdom Tools to test the system's functionality, the test results showed that the original system cannot meet the management needs of users for competitive competitions and physical fitness testing. However, this system fills this gap and provides convenience for the work and learning of teachers and students. In order to solve the problem of time and space constraints

in physical education teaching, Liu [7] designed a physical education teaching system based on the actual needs of physical education teaching. The framework was designed using MVC mode, and the system data processing logic was implemented in layers. The teaching content was sorted and classified from multiple aspects such as user login, file management, score management, courseware release, system maintenance, and teacher–student communication, achieving independent learning and free exploration. The application of physical education teaching system can not only improve students' learning enthusiasm, but also achieve the goal of alleviating teaching pressure and improving teaching efficiency. Chen and Dou [8] designed and built an intelligent physical education teaching system in colleges and universities based on the knowledge system, and made a detailed analysis of the functions of the system modules, namely user management, network course management, question database maintenance management, real-time examination management, grading audit management, system management, user health information collection, and database design. Research has shown that this system can not only significantly improve the efficiency and quality of physical education theory teaching, but also fully mobilize students' learning enthusiasm and initiative, to promote students' healthy and stable development. Zhao [9] proposed a design of an intelligent physical education teaching system based on the dynamic characteristics of physical education teaching. In application, it can be found that the design of a knowledge-based intelligent physical education teaching system is flexible in teaching, not limited by time and location, and can meet the needs of students in different situations. To ensure the security and integrity of physical education and user systems, Su and Zhou [10] first needed to use advanced and secure backup systems to preserve the integrity of data. In addition, the mobile sports teaching network system should adopt a standard and standardized online system to ensure the security of users and information, and update data content in a timely manner to ensure that the information users see is timely and accurate. The entire data of the system should be encrypted to improve the security and stability of the system. Physical education is already widely acknowledged as being crucial for students. Students' mental health can be improved through physical education in addition to their physical health. They have carried out various experiments to prove the importance of sports, but no detailed experimental data.

At present, the agent has been widely used in e-commerce, spacecraft control, distance education and other fields. The agent system is an important part of artificial intelligence, and the multi-agent system focuses on the cooperation mechanism between the two agents. Nie et al. [11] found that agricultural decision-making received increasing academic attention. His purpose was to analyze subjective

and objective risk preferences, and establish a multi-agent model. Multi-agent was used to simulate the actual agricultural decision-making of farmers. Dey et al. [12] argued that traditional temperature management techniques were not smart enough to meet system performance. To solve the integrity tracking problem, Zhao et al. [13] introduced the innovative concept of multi-agent. Under the concept of multi-agent, consensus tracking protocol has been unified to ensure the reachability of sliding motion in a limited time. Yu et al. [14] first proposed a new split distributed sliding mode control for the second-order consistency of multi-agent systems, and finally solved the basic unknown problem of sliding mode control design for coupled network systems. He designed a proxy system to remove the effects of the singularity problem. Scholars have already known that the multi-agent model can be applied to many fields, and the multi-agent mode system has higher performance than the single-agent model system. Therefore, it is advantageous to apply multi-agent model to the design of the system for teaching physical education to enhance system performance.

The system designed in this paper matched the best sports training items and programs according to the dynamic changes of users' sports ability level. Finally, through application analysis, the system can effectively improve users' perceived ease of use and effectiveness experience, and can meet the personalized sports training needs of different users. The multi-agent system can provide personalized teaching service according to different needs and abilities of students. Each student can have an independent agent, according to their learning progress and characteristics of intelligent guidance and assessment, to help students better development and learning.

3 Design of Sports Teaching Management System Based on Multi-agent Model

Because physical education requires a high degree of interaction and practice between teachers and students, the implementation of physical education network teaching is generally more demanding than other disciplines. The importance of social and economic development provides a good opportunity for online physical education, but the realization of online sports still depends on the rapid development of information technology [15]. The foundation for implementing online instruction in physical education is provided by the development of e-books, multimedia courses, and online video playback platforms. Online sports platforms have progressed and grown more affordable with the use of sophisticated multimedia technologies, giving students a positive learning experience and the convenience of online learning. The establishment of a networked physical education teaching approach is essential given the backdrop

of national college teaching reform and the development of social science and technology.

3.1 Generation Method of Multi-agent Resource Demand

Multi-agent is a large complex system loosely coupled by multiple problem solvers distributed on the network, and these problem solvers interact to solve complex problems that cannot be handled by a single individual's ability knowledge. Agent refers to a computing entity with habitable, reactive, social and program characteristics that exists in a specific environment and can continue to work autonomously [16]. An example of agent feature transfer learning is shown in Fig. 1.

As shown in Fig. 1, transfer learning can transfer the knowledge acquired from experts or other processes to the current task to accelerate learning. Transfer learning has been widely studied in the field of supervised learning. Migratability can quickly learn new complex tasks by automatically creating empirical knowledge or models. The goal of transfer learning is to find and exploit the correlation between different learning tasks, and use past learning experience to accelerate current and future learning. Virtually every advanced learning and memory process involves transfer, which often allows all types of learning processes to go beyond simple stimulus–response cycles [17]. For different scheduling models, there are two relationships between construction period and resource demand: construction period has nothing to do with resource demand, and construction period decreases monotonically with resource demand. Therefore, before assigning resources to task execution patterns, the relationship between individual resources and

duration should be determined. The relationship between the duration of work and the amount of resources can be expressed in Formula 1.

$$F_m(r) = P_m. \quad (1)$$

If $F_m(r) = 1$, it means that the duration and resources of the pattern are two independent variables. Then, the demand for this resource of each job, and the generated random resource demand are allocated to each execution model of the job that needs the resource [18].

The mode can be divided into M_j execution modes with different durations as Formula 2:

$$\Delta = \frac{U^{\text{high}} U^{\text{low}}}{M_j}. \quad (2)$$

According to different resource intervals, the resource demand in the corresponding model can be generated as Formula 3:

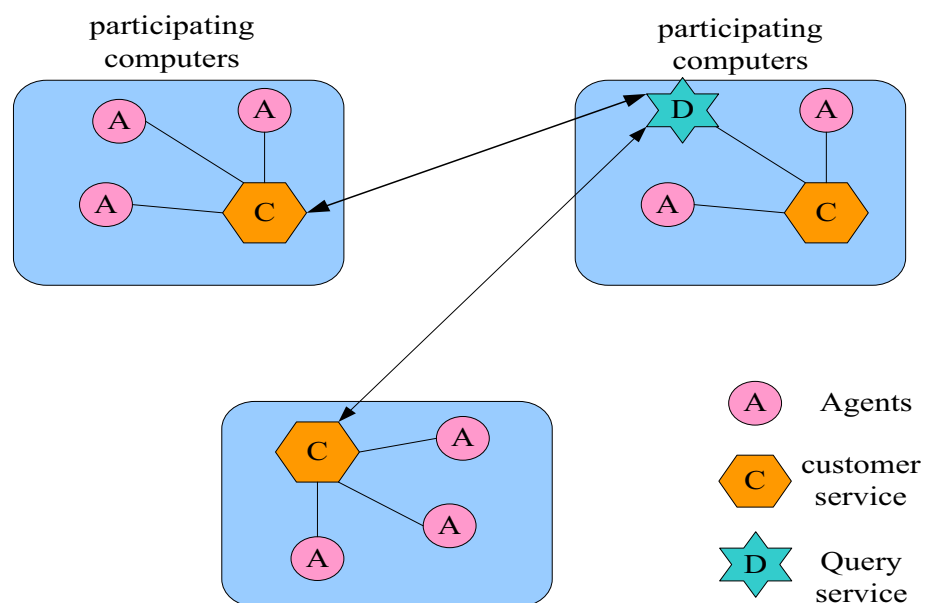
$$I_k = [\text{round}(U^{\text{high}} - \Delta k), \text{round}(U^{\text{high}} - \Delta(k - 1))]. \quad (3)$$

This paper introduces the concept of resource intensity as Formula 4. If the network plan arranges multiple activities to be carried out simultaneously in a certain period of time, the amount of a certain kind of resource (called resource intensity) required per unit time is:

$$RS_r = \frac{K_t}{\frac{1}{J} \sum_{i=1}^J k_{jr}}. \quad (4)$$

To facilitate the solution of a single resource stock, this paper regards resource intensity as the resource stock and

Fig. 1 Example of agent feature transfer learning



the conversion factor between the minimum and maximum resource stock [19]. The inventory of resources in the system is solved to Formula 5 using the following formula.

$$K_t = K_r^{\min} + RS_t(K_r^{\max} - K_r^{\min}), \quad (5)$$

For a non-renewable resource r , the minimum and maximum supply of the resource is solved by Formula 6.

$$K_t^{\min} = \sum_{j=2}^{J-1} \min \{K_{jm}^v\}. \quad (6)$$

For renewable resources, the minimum and maximum resource supplies are solved using Formula 7.

$$K_t^v = K_t^{\min} + \text{round}(RS_t(K_r^{\max} - K_r^{\min})). \quad (7)$$

In the negotiation process between different types of agents, certain behavioral rules and algorithms are needed to help the agents make more reasonable decisions, so as to better optimize the global performance while satisfying local interests [20]. Good behavior rules can help agents make accurate decisions and greatly improve the efficiency of cooperation between agents.

In the process of project planning, all working agents are first stored in an initial data structure, and constantly browse the system information. When information about priorities and resource requirements is available, work is scheduled according to the chosen execution model [21].

3.2 Multi-agent Decision Rule Design

The main goal of multi-agent is to optimize the success rate of system scheduling while minimizing the consumption of non-renewable resources by unplanned work according to predetermined decision rules [22]. The schematic diagram of multi-agent decision rule design is shown in Fig. 2:

As shown in Fig. 2, the multi-agent decision rule design process includes problem determination, agent identification, decision rule design, evaluation and deployment. It is necessary to define the problem and goal to be solved, the scope, constraints and evaluation indicators of the problem, and determine the agents involved in solving the problem and their characteristics, operations and responsibilities. The performance and effectiveness of models and decision rules are evaluated using set evaluation metrics, scenarios, or data sets, and optimized models and decision rules are deployed to real application scenarios, so that they can truly solve problems and achieve goals.

The estimation of the possible demand for non-renewable resources is the basis of many resource agent decision guides. It is calculated as Formula 8:

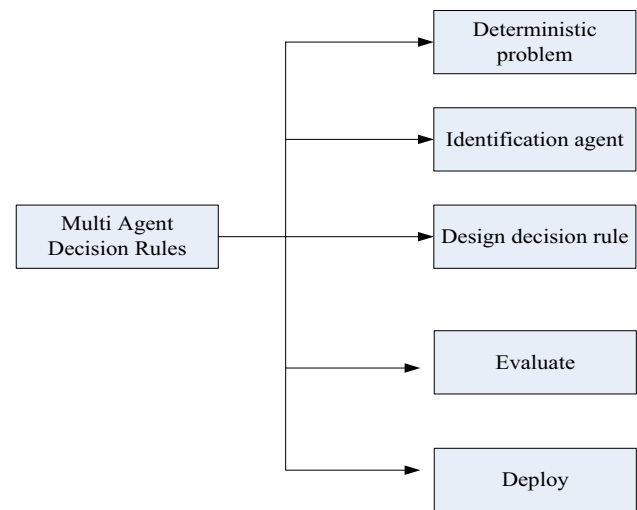


Fig. 2 Schematic diagram of multi-agent decision rule design

$$K_t^{\min} = \sum_{j=2}^{J-1} \min \{K_{jm}^v\}. \quad (8)$$

The decision rule of resource agent basically needs to design a specific algorithm. Through the mode agent, the minimum priority rule for non-renewable resource requirements is invoked. This maximizes the use of non-renewable resources by the system and guarantees the success rate of system scheduling [23]. Resource agent allocation rules algorithm is an algorithm used to manage and allocate resource agents to ensure that resources can be efficiently and fairly allocated to different agents. Pseudocode is an informal English-like structure used to describe block figures. Formula 9 is the pseudocode of the resource agent allocation rule algorithm:

$$\sum_{n=1}^N k_{jmn}^N = \min \left\{ \sum_{n=1}^N k_{jmn}^N \forall n : \text{PFDNR}_n > K_n^N \right\}. \quad (9)$$

The calculation method of the agent's decision function in the improved system is Formula 10:

$$\text{SchedJob} = \frac{S}{T}. \quad (10)$$

In Formula 10, S represents the number of immediate successors of the current job. If the value of is larger, the work would be correspondingly higher [24].

This method represents the intelligence of the agent and the optimal performance of the scheduling system. It is Formula 11:

$$H = \left(\sum_{r=1}^R k_{jmr} + \sum_{r=1}^R k_{jmn} \right). \quad (11)$$

This technique fundamentally adopts the concept of holistic model to study knowledge transfer. The reusability of the model is examined, and the source domain model is taken as the prior knowledge of the target domain. The knowledge contained in the model is augmented with prior knowledge, and information transfer occurs in the target domain [25]. SVM generally refers to support vector machine. Support vector machines are a class of generalized linear classifiers that perform binary classification of data in a supervised learning manner. By using bias regularization, the model parameters trained for the SVM in the source domain can be moved to the target domain, as shown in Formula 12:

$$K = \min \frac{1}{2} \|w - w_0\|_2^2 + \lambda \sum_{i=1}^l \xi_i. \quad (12)$$

Formula 12 represents the successfully trained model parameters. The same idea is also reflected in the maximum posterior probability method:

$$G = \max P_r(w|w_0) P_r(\{(a_i, b_i)\}_{i=1}^I | w). \quad (13)$$

In this paper, the separation prior and target classifier are predicted from a new Bayesian perspective, and the conclusion of a partially adaptive boundary is drawn. It is also found that the model-based transfer learning framework can be extended to unlabeled data in the target domain.

This paper proposes a new challenge for multi-task learning-learning framework. To make it better understood, this paper assumes that there are two fields, and the obtained model parameter is Formula 14:

$$\tilde{w} = w_0 + \tilde{w}_\Delta. \quad (14)$$

Knowledge transfer optimization variables w_0 and $\tilde{w}_\Delta$ represent optimization variables under the SVM framework, which can be performed simultaneously and play a role.

Among them, the calculation method of the current resource factor is Formula 15:

$$ARF_r = \frac{1}{J-2} \frac{1}{|\tau|} \sum_{j=1}^{J-2} \frac{1}{M_j} \sum_{m=1}^{M_j} \sum_{r \in \tau} Rq[j, m, r]. \quad (15)$$

Since physical education teaching and physical education management are in a proportional relationship, it is necessary to rank physical education teaching in the management work, and allocate the corresponding resource requirements according to the order of physical education teaching, as Formula 16:

$$U^{\text{low}} = \min [U^1, U^2]. \quad (16)$$

When the number of demand types for this type of resource in the working mode combination is less than the maximum number of resource demand types, the value of the current resource factor is calculated according to the currently selected resource type, and the current resource factor is compared with the resource factor defined by the system. Then different working model combinations are randomly selected from the set CT until the resource factor defined by the system is reached.

3.3 Design of Physical Education Teaching Management System

If only one agent is created, a large amount of uncertain information can only be processed by one agent, which is easy to become illogical. Multi-agent can successfully complete tasks through task division, the creation of various agents, and the communication and cooperation between agents. Each agent can share certain responsibilities and have specific roles in this way, thereby improving the overall efficiency of the system and preventing agent confusion [26, 27].

The multi-agent development platform JADE uses MyEclipse8.0 as the development environment of the multi-agent-based physical education management system. It includes a running environment on which agent lives, a class library for developing agent applications and a set of graphical tools for debugging and configuration, which simplifies the development process of a multi-agent system. Function JADE provides many functions for multi-agent systems. The benefit of MyEclipse 8.0 is that all components can be successfully installed at once without testing. The integrated development environment of MyEclipse is shown in Fig. 3.

The database is a key component in creating an information system, as shown in Fig. 3. After the system database is created, the database can be accessed.

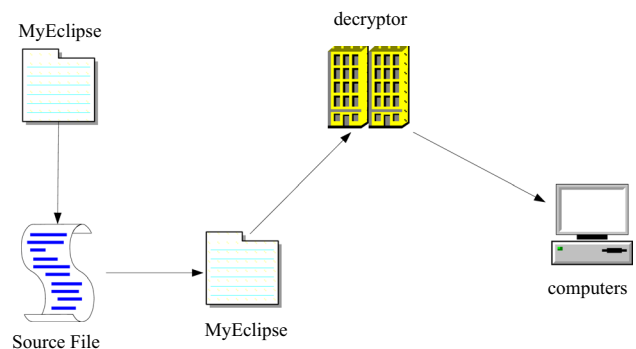


Fig. 3 MyEclipse IDE

The reservation information management sub-module is one of the key elements of the system, which realizes the dynamic management of reservation information for the convenience of users. MyEclipse is the most comprehensive Java IDE (integrated environment), and MyEclipse provides intelligent enterprise-class tools for development tasks. Most of the existing online teaching activities adhere to the teaching process of “register and log in—teach—leave the system”. A single approach cannot solve all the problems in this process. In Fig. 4, the entire model architecture is shown.

As shown in Fig. 4, in this model paradigm, the user layer, the middle agent and the server database layer are used horizontally. The main function of the middle agent layer is to realize the multi-agent communication language for the communication between agents. The used of this setting can significantly improve access speed. Users can use a browser to access. Today, more and more college students use smartphones, most of which provide internet access [28, 29].

1. User interface agent

The main responsibility of the agent is to provide user registration and login services. Based on the user's registration details, the legitimacy of the user can be determined when they log in. If the username and password are accurate, the relevant student or teacher agent would be generated and various interfaces would be provided for users with various login IDs. If the student is a newly registered user, the registration details in the User Basic Information Data Sheet must be updated as the basis for subsequent services. If a student completes their coursework and leaves the system, the student's learning record would be saved in the Learning Records Agent [30].

2. Q&A Agent

Student troubleshooting is the goal of the Q&A service. This paper shows that one of the main functions of question answering agent in the basic design of the system is to

respond to common queries. To realize question answering, question answering agent cooperates with student agent to solve students' problems. If students still cannot solve the problem after learning the necessary materials, they can ask the Q&A Agent for help. However, the question and answer database created by teachers may not be completely accurate, and students can currently seek help through forums [31, 32].

3.4 Realization of Intelligent Agent Module

With the rapid development of computer technology, intelligent agent technology is becoming more and more popular. It has been used in a variety of fields, including intelligent learning guidance, network management, and air traffic control. According to the requirements of the system, this paper chooses Jade as the multi-agent development platform [33].

In a centralized management system, decision-making power is usually concentrated in the hands of a central organization or individual, and the system has high stability. The decision-making process is relatively simple and the allocation of resources is efficient, but it may lack flexibility and innovation.

In a decentralized management system, power and responsibility are dispersed to various levels of institutions or individuals, and this system has greater flexibility and innovation. Each management unit can make decisions according to the actual situation, but it may also lead to inconsistent decisions or uneven resource allocation. The challenge and focus of this paper is how to create each agent module and coordinate its interactions according to the unique capabilities of each intelligent agent. By placing intelligent agent modules in key positions, not only the stability and security of the system are improved, but also the management workload is reduced. The intelligent agent module is shown in Fig. 5.

As shown in Fig. 5, both the login module and the subscription sub-module can be implemented using code when creating the management system. However, these two system sub-modules use intelligent agent technology, which greatly improves the intelligence of the system. Agent technology is a computer system packaged in a certain environment. To achieve the design purpose, it can be flexible and autonomous in this environment. It is realized through the cooperation of session agent and database agent. Especially in the reservation sub-module, the session agent must receive the user's reservation information, and then must send it to the database agent. Database agent obtains retained data, provides analysis and performs auditing of retained data. Each agent performs the same function as a single person, and each agent has a specific role. A number of agents are introduced into the system, and each agent has its own unique set of responsibilities.

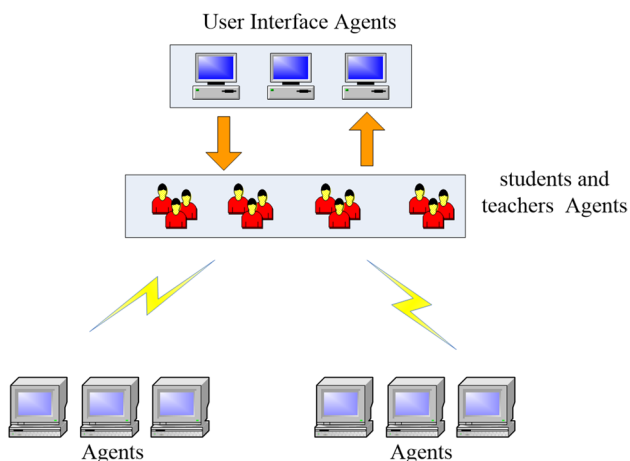


Fig. 4 Basic framework of multi-agent-based teaching management system

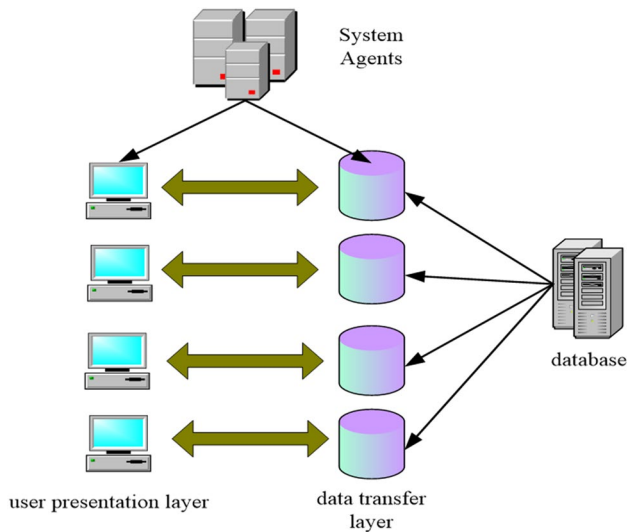


Fig. 5 Intelligent agent module

Intelligent agents need to have the ability to perceive the environment and obtain information about the state of the environment through sensors or other means. The perception module can include functions such as sensor data collection, signal processing, and data processing.

The performance of the physical education management system based on multi-agent is obviously better than the

traditional management system. Traditional systems development methods are not fast enough to process appointment data. For example, the administrator may be sleeping or taking a day off and cannot decide the user's appointment information in time. After joining an agent, this is the same as appointing a 24-h administrator, which constantly checks all aspects of the system and completes any tasks requested by the user.

4 Development of Teaching Management System and Performance Test Experiment

4.1 Development of Physical Education Management System

The development of online physical education in China has achieved initial success, although professional institutions and colleges are currently the focus of development. Online teaching is developing rapidly, especially in some sports schools and universities. Famous colleges and universities have developed sports network teaching platforms to provide students with rich multimedia courseware. The development trend of physical education teaching from 2016 to 2019 is shown in Fig. 6.

As shown in Fig. 6, the growth rate of schools that established their own teaching network in 2016 and 2017

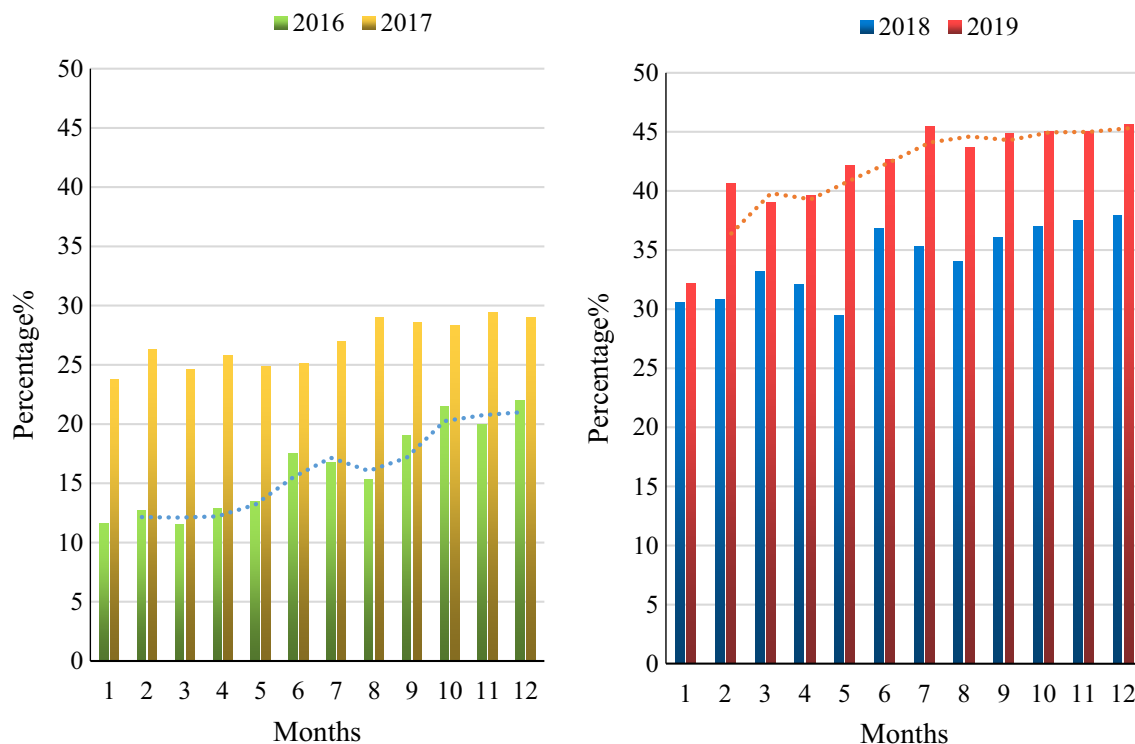


Fig. 6 Development trend of physical education teaching from 2016 to 2019

Table 1 Basic information of 300 students

Index	Index	Number of people	Percentage (%)
Gender	Male	200	66
	Female	100	34
Age	18–21	150	50
	22–25	150	50
Have a mobile device	Yes	270	90
	No	30	10

was lower than that of schools that established their own teaching network in 2018 and 2019. With the help of the government and funds, major colleges and universities across the country have established their own teaching networks, and some have even successfully taught some core courses online. Even within schools, there are many distance learning platforms that make the most of existing teaching materials. With the increasing popularity of network teaching, college sports network teaching has been popularized and developed rapidly.

This paper conducted a survey on 300 physical education students at Yulin Normal University, and their basic information is shown in Table 1.

As shown in Table 1, the hardware foundation of mobile learning is mobile devices. According to the report, all college students own a mobile device, about 90% of which are smartphones. This may be due to the current relatively high living standards per capita. On the other hand, it is also due to the increase in smartphone usage and falling costs. With the widespread use of smartphone devices, the gap between resource display and standard computer display is gradually narrowing, creating a better environment for mobile learning. In addition to watching videos, students can also learn through text materials. In addition, mobile learning has a broader space for development, because learners can also learn through specific learning websites.

This paper then investigated and analyzed whether students were satisfied with the traditional teaching management system, as shown in Table 2.

Table 2 demonstrated that 30 persons, or 10% of the population, were highly satisfied with the current instructional management system. A total of 33 users, or 13% of the population, were happy with the existing teaching management system. There were 60 users, or 20% of the population, who were usually satisfied with the existing teaching management system. A total of 87 people, or 29% of the population, were unhappy with the existing educational management system. A total of 90 people, or 30% of the population, were extremely unhappy with the system. It is clear that the

Table 2 Student satisfaction survey on the traditional teaching management system

Satisfaction level	Number of people	Percentage (%)	Effective percentage (%)
Very satisfied	30	10	10
Satisfied	33	11	11
Generally	60	20	20
Dissatisfied	87	29	29
Very dissatisfied	90	30	30

majority of students believe the existing educational system is not very effective.

It demonstrates that there are still some issues with the acceptance and application. This paper invited ten experts to rate it, as shown in Fig. 7.

As shown in Fig. 7, many teachers do not understand the concept of online teaching, so the process of teacher team building in online teaching needs to be improved. One of the main goals of promoting online teaching is to speed up the transformation of teachers' roles, help teachers break away from traditional teaching as soon as possible, and master the methods of online teaching. At present, there are few high-quality materials for physical education, which is another weakness of sports network teaching. People must speed up the construction of efficient physical education courses and improve the standards of sports online teaching materials. In addition, through the analysis of sports network teaching evaluation, it is believed that the current evaluation system cannot stimulate people's enthusiasm.

Although the maintenance of the teaching management information system involves individuals who are familiar with the system functions, most users who use the system in colleges and universities are not professional computer personnel, and their computer operation level is usually not high. In this sense, colleges and universities lack full-time system management skills. Even if they do, their work environment do not prioritize system administration, making it impossible for them to concentrate on maintaining instructional management information systems.

4.2 Physical Education Management System Test

System testing is a way to ensure that a system is implemented correctly. To ensure that there are as few bugs as possible in small functional modules in the system and to reduce the burden of later completion, continuous testing must be performed during system development. The experimental equipment for this paper used a later CPU and equipment with a monitor, keyboard, and mouse, etc.

Three students were selected to conduct system stability experiments on the traditional teaching management

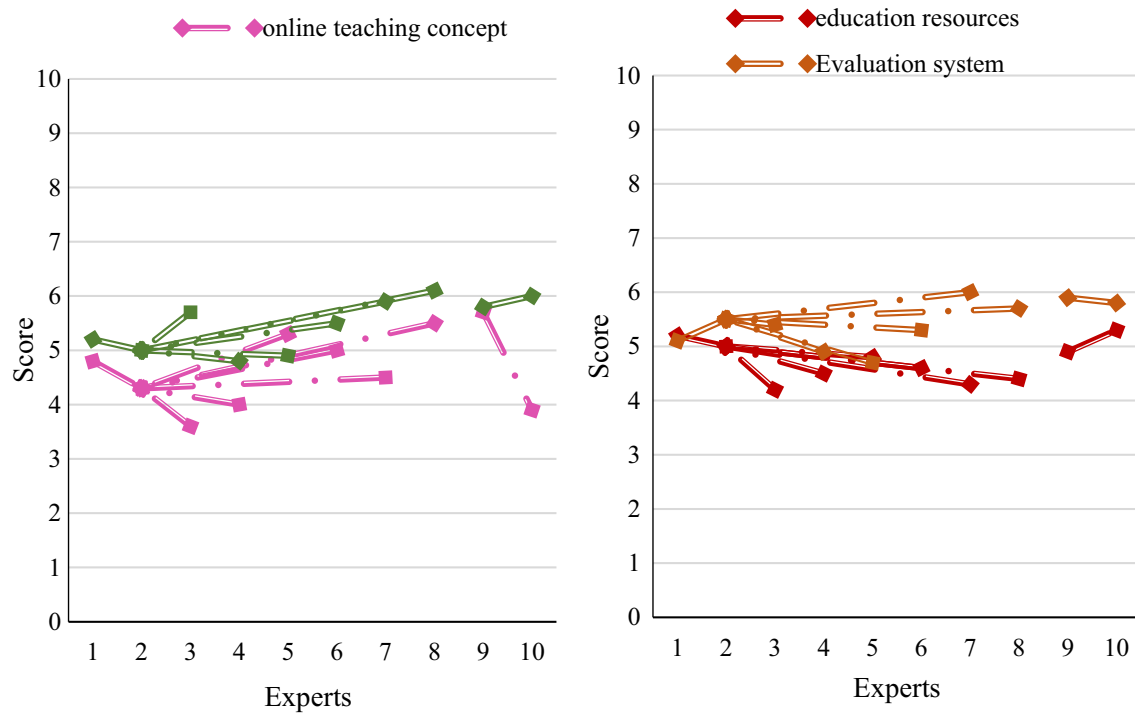
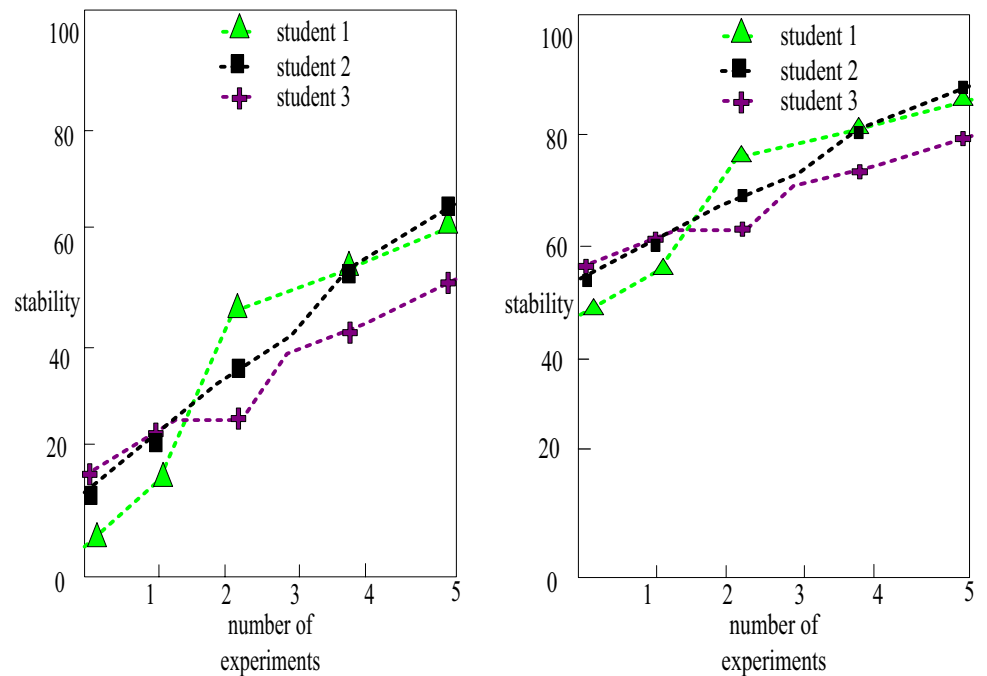


Fig. 7 Problems existing in the current teaching management system

Fig. 8 Comparison of stability of different management systems



(a). Stability of traditional teaching management system

(b). Stability of teaching management system based on multi-agent mode

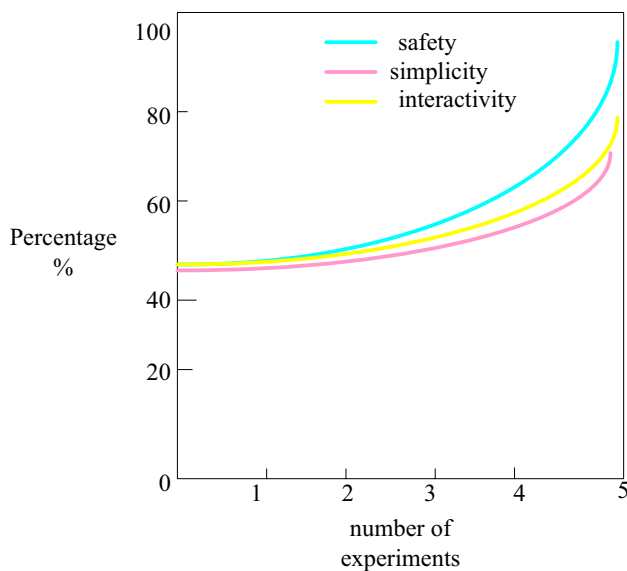


Fig. 9 Main features of the teaching management system in this paper

Table 3 Comparison results of five experiments

Number of experiments	Traditional system		Multi-agent system	
	Responding speed (s)	Efficiency (%)	Responding speed	Efficiency (%)
1	1.2	70	0.2	93
2	1.1	73	0.1	95
3	1.3	69	0.3	90
4	1.4	67	0.5	88
5	1.0	75	0.2	93

system and the system developed in this paper. The experimental results are shown in Fig. 8.

According to Fig. 8, the stability of the conventional teaching management system was shown in Fig. 8a to be approximately 65% at the highest and 10% at the lowest. Figure 8b showed that the multi-agent mode-based teaching management system's quality ranged from about 90% at the highest level to about 50% at the lowest. The system created in this research has more stability than the conventional management system for physical education.

Figure 9 displays the key characteristics of the teaching management system created in this study.

As shown in Fig. 9, the system not only reduces the cost and cycle of system development, but also breaks the limitations of the original teaching management system, ensuring the safety of key data, the speed of operation, simplicity and convenience, and the benign interaction of the system. It also reduces maintenance costs.

This paper compared the traditional teaching management system with the system proposed in the article. The comparison results of the five experiments are shown in Table 3:

As shown in Table 3, the system has high operational efficiency and fast query and retrieval speed. Through experimental research and related practical applications, it is demonstrated that the approach totally overcomes the shortcomings of conventional college physical education instruction, allowing for greater flexibility in physical education and learning that is neither time- or location-based. It significantly increases student engagement and learning efficiency and quickly raises the bar for physical education in colleges and universities.

5 Conclusions

Through the collaboration and interaction between multi-agents, the system can provide personalized teaching support according to the individual differences and learning needs of students. Each student can have an independent agent who can provide targeted learning resources, guidance and assessment based on the student's abilities, interests and learning progress, enabling students to learn at their own pace and in their own way. Physical education management system can provide teachers with powerful guidance and monitoring tools. Teachers can use agents to monitor students' learning progress and performance, and to identify and solve students' difficulties in a timely manner. In addition, teachers can also understand students' learning habits, problems and needs through the interaction between agents and data analysis, so as to provide them with personalized teaching guidance. The conventional management system for physical education had several issues, including weak stability, low efficiency, and poor safety. Therefore, this article suggested using the multi-agent model in the system's design and development to increase the efficiency of the teaching system. This study modified the system's design by incorporating an intelligent multi-agent model, which not only raised the level of human-machine interface contact but also quickly responded to student inquiries, enhancing the system's management effectiveness. It was discovered that the majority of students were dissatisfied with the current system, making the development of a new system essential. However, because this paper's experimental resources were constrained, the experiment had numerous flaws. The scientificity and rigor of the experiments should be strengthened in future work. The physical education teaching management system based on multi-agent models can achieve more intelligent, personalized, and collaborative teaching management services in the future, improving students' learning effectiveness and teachers' teaching quality. Meanwhile, with the continuous development of artificial intelligence and data

analysis technology, more innovations and applications would emerge in this field.

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Data Availability Relevant data for this work are available to the extent reasonably requested by contacting the corresponding author.

Declarations

Conflict of interest We were able to confirm that there was no conflict of interest in this work.

Institutional Review Board Not applicable.

Informed consent Not applicable.

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